

A Layered HFSLs-PSO-BIGRU Framework for Stock Forecasting and Trading in the Taiwan Stock Market

Po-Chun Huang^{1,*} and Hung-Wen Cheng¹

¹Department of Data Science, Soochow University, Taipei, Taiwan

Abstract

Stock return forecasting in high-dimensional, nonlinear, and non-stationary markets remains a core challenge in quantitative finance, and deep-learning models calibrated on a single market often fail to generalize. This study applies a layered framework integrating Hierarchical Feature Selection with Local Shuffling (HFSLs), Particle Swarm Optimization (PSO), and Bidirectional Gated Recurrent Units (BIGRU) to the Taiwan stock market, using daily Alpha158 features from January 2020 to September 2025 for the TAIEX and four sector-representative stocks—TSMC, Evergreen Marine, MediaTek, and Uni-President. Ablation experiments show that HFSLs is the dominant driver of performance, raising R^2 by 2.392 and cutting MSE by 97.1% relative to the baseline BIGRU, while PSO applied alone destabilizes learning; the full HFSLs-PSO-BIGRU model attains an average R^2 of 0.902 across the five targets. Translating the eleven-day forecast into a symmetric long-short strategy with an eleven-bucket capital-allocation scheme to resolve overlapping trades, the model delivers cumulative returns and profit factors that substantially exceed both the BIGRU baseline and the TAIEX benchmark on all four stocks, evidencing robust cross-market and cross-sector generalization.

Keywords: layered stock prediction; hierarchical feature selection; particle swarm optimization; Taiwan stock market; quantitative trading

JEL Classification: G11; G17; C45; C53; G10

1 Introduction

The continuous advancement of financial technology has positioned machine learning and deep learning as central tools for stock return prediction. Traditional technical analysis and econometric models typically rely on a limited number of indicators and linear assumptions, which restrict their ability to capture the complex, nonlinear dynamics observed in real markets (Fama and French, 1992; Malkiel, 2003). Empirical studies show that machine learning methods significantly improve asset return predictability relative to traditional benchmarks (Gu et al., 2020), and deep learning architectures are particularly effective at modeling the nonlinear relationships and temporal dependencies inherent in financial time-series data (LeCun et al., 2015).

Among deep learning approaches, recurrent neural networks and their variants are widely adopted for time-series forecasting: the LSTM addresses the vanishing-gradient problem through gating (Hochreiter and Schmidhuber, 1997; Bengio et al., 1994), the GRU offers a parameter-efficient alternative (Cho et al., 2014), and bidirectional structures incorporate both past and future context (Schuster and Paliwal, 1997). Nevertheless, on high-dimensional feature sets these models are prone to the curse of dimensionality and overfitting, degrading out-of-sample performance (Guyon and Elisseeff, 2003; Hastie et al., 2009); in finance, improper validation can further produce strategies that suffer from backtest overfitting (Bailey et al., 2014; López de

*Corresponding author. Tel.: +886-989871776
E-mail address: a08284700711@gmail.com.

Prado, 2018). To address these challenges, this study integrates a hierarchical feature selection mechanism (Li et al., 2017) for redundancy reduction with particle swarm optimization for hyperparameter tuning (Kennedy and Eberhart, 1995; Bergstra and Bengio, 2012), applied to a bidirectional GRU sequence model.

Although this layered framework was originally validated on the Chinese market, whether it retains its predictive accuracy and stability on the Taiwan market—which differs substantially in microstructure, regulation, and investor composition—has not been empirically established.

This paper makes three contributions: (1) a systematic cross-market evaluation of the layered HFSL-PSO-BIGRU framework on Taiwan data, covering both the market index and four sector-representative stocks; (2) ablation experiments quantifying each component’s marginal contribution and the conditions under which PSO helps; and (3) a symmetric long–short trading strategy with a capital-allocation scheme that resolves the overlapping-trade problem of multi-day-ahead forecasts, benchmarked against passive holding and the TAIEX.

The remainder of the paper reviews related work (Section 2), presents the framework (Section 3) and experimental setup (Section 4), reports prediction and ablation results (Section 5) and the trading strategy and backtests (Section 6), and concludes (Section 7).

2 Related Literature

Traditional statistical methods. The ARMA model developed by Box and Jenkins (Box and Jenkins, 1970) captures linear dependencies between observations and error terms, and was later extended to ARIMA for non-stationary series (Box and Jenkins, 1970). To model volatility clustering, Engle (Engle, 1982) proposed the ARCH model, generalized into GARCH by Bollerslev (Bollerslev, 1986). These models rely on assumptions of linearity and stationarity that are often violated in real markets with nonlinear dynamics and structural breaks (Hastie et al., 2009), limiting their performance in complex, high-dimensional environments.

Machine learning methods. Support Vector Machines (Vapnik, 1995), Random Forests (Breiman, 2001), and other nonparametric approaches capture nonlinear relationships among variables but remain susceptible to overfitting in high-dimensional settings (Guyon and Elisseeff, 2003; Hastie et al., 2009) and largely ignore temporal dependencies. Recent studies show that machine learning significantly enhances asset pricing and return prediction (Gu et al., 2020; Kelly et al., 2019); statistical-arbitrage work on the S&P 500 confirms that deep networks, gradient-boosted trees, and random forests beat traditional benchmarks (Krauss et al., 2017). With more than 240 candidate return predictors documented (Harvey et al., 2016), combining advanced models with robust feature selection is increasingly important.

Deep learning methods. Deep models capture complex nonlinearities through hierarchical representation learning (LeCun et al., 2015). RNNs suit sequential data but suffer from vanishing gradients (Bengio et al., 1994); LSTM addresses this via gating (Hochreiter and Schmidhuber, 1997; Greff et al., 2017), GRU simplifies LSTM for efficiency (Cho et al., 2014), and bidirectional RNNs add backward context (Schuster and Paliwal, 1997). Attention and Transformer architectures further improve long-range dependence modeling (Vaswani et al., 2017; Lim et al., 2021), and LSTM-based models have been applied to large-scale market forecasting and trading (Fischer and Krauss, 2018). Yet deep models remain sensitive to redundant high-dimensional inputs (Goodfellow et al., 2016), making feature selection and hyperparameter optimization essential.

3 Methodology

The proposed HFSL-PSO-BIGRU framework adapts the layered model of Chen and Tang (2025) to the Taiwan market, comprising normalization, hierarchical feature selection with local shuffling, a bidirectional GRU predictor, PSO-based hyperparameter tuning, and multi-layer weighted aggregation (Fig. 1).

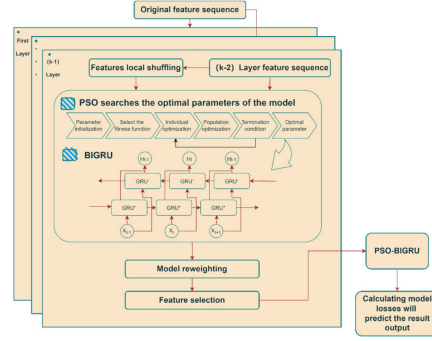


Figure 1: The layered HFSLS-PSO-BIGRU model framework. Adapted from Chen and Tang (2025).

3.1 Data Preprocessing and Normalization

Stock market data contain multi-dimensional features whose value ranges vary significantly. To prevent instability during training, all features are normalized to $[-1, 1]$ as $x_{\text{norm}} = (x - x_{\min}) / (x_{\max} - x_{\min})$, where $x_{\min}$, $x_{\max}$ are the feature minimum and maximum; this accelerates convergence and improves performance.

3.2 Hierarchical Feature Selection with Local Shuffling (HFSLS)

Traditional feature selection methods often require retraining the model to evaluate feature importance, incurring high computational cost. HFSLS instead assesses importance by perturbing individual features and measuring the resulting change in prediction loss, without retraining. The original and feature- i -perturbed prediction errors are $L = (\bar{M} - y)^2$ and $L_i = (\bar{M}_i - y)^2$, where $\bar{M}$ ($\bar{M}_i$) is the mean prediction on the original (perturbed) data and y is the true label. The feature importance is then

$$E_i = \frac{\sigma(L_i - L)}{E(L_i - L)}, \quad (1)$$

where E is the expectation operator and σ the standard deviation; a large E_i indicates high importance. In the hierarchical framework, each layer selects features from the previous layer using local shuffling and sampling, ensuring the feature count decreases layer by layer ($|F_1| > |F_2| > \dots > |F_K|$), reducing redundancy and complexity.

3.3 Bidirectional Gated Recurrent Unit (BIGRU)

To capture the temporal dynamics and contextual dependencies in stock price time series, the framework employs BIGRU as the core predictor. The GRU mitigates the vanishing gradient problem of traditional RNNs through update-gate (z_t) and reset-gate (r_t) mechanisms. The candidate hidden state and hidden-state update are

$$\tilde{h}_t = \tanh(W_{\text{ih}}x_t + W_{\text{hh}}(r_t \odot h_{t-1}) + b_h), \quad (2)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t. \quad (3)$$

The BIGRU combines forward and backward sequences to capture the complete temporal context, and the final output concatenates the two hidden states, $h_t = \text{concat}(h_t^{\text{forward}}, h_t^{\text{backward}})$.

3.4 Particle Swarm Optimization (PSO) for Hyperparameter Tuning

Network performance depends heavily on hyperparameters (GRU units, dropout rate, learning rate); manual tuning is inefficient and prone to local optima, so PSO searches for the global optimum. Each particle updates its velocity, inertia weight, and position by

$$\begin{aligned} v_i(t+1) &= \omega v_i(t) + c_1 r_1 (P_i^{\text{best}} - x_i(t)) + c_2 r_2 (P_g^{\text{best}} - x_i(t)), \\ \omega &= \omega_{\max} - \frac{n}{n_{\max}} (\omega_{\max} - \omega_{\min}), \quad x_i(t+1) = x_i(t) + v_i(t+1), \end{aligned} \quad (4)$$

with fitness given by the mean squared error $\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$. Particles iterate until reaching the maximum number of iterations or the global best hyperparameter set.

3.5 Multi-Layer Integration and Weighted Aggregation

In the hierarchical framework (Fig. 1), each layer generates an individual prediction from progressively refined feature subsets. Let k be the number of layers and $\hat{y}_i$ the prediction from the i -th layer. The final prediction is

$$\hat{y} = \sum_{i=1}^k w_i \cdot \hat{y}_i, \quad w_i = \frac{1 + (i-1)s}{k + \frac{k(k-1)s}{2}}, \quad i = 1, 2, \dots, k, \quad (5)$$

where s is the weighting step controlling the contribution difference between layers; the weights are normalized and sum to one. When $s = 0$ the scheme reduces to uniform averaging ($w_i = 1/k$), which experiments show provides the most stable performance; for $s > 0$, higher layers receive larger weights, reflecting their refined feature representations.

Throughout, the target return is the logarithmic return $R_{t+h} = \ln(S_{t+h}/S_t)$ —additive, scale-independent, and symmetric in up and down moves—where S_{t+h} , S_t are the closing prices at times $t+h$ and t .

4 Data and Experimental Setup

4.1 Data Sources

The dataset is sourced from the Taiwan stock market via the TEJ Pro database, comprising daily closing prices from January 1, 2020 to September 2, 2025. To avoid information leakage, it is split chronologically: training (2020-01-01 to 2023-09-30), validation and hyperparameter tuning (2023-10-01 to 2024-03-31), and the out-of-sample test set (2024-04-01 to 2025-09-02). The primary target is the TAIEX; to evaluate cross-industry generalization, four representative stocks are added: TSMC (semiconductor), Evergreen Marine (shipping), MediaTek (IC design), and Uni-President Enterprises (retail). The Alpha158 feature set—158 technical indicators spanning price trends, momentum, and volatility—is computed programmatically from raw price and volume data rather than from pre-built indicators, ensuring consistency and reproducibility.

The Alpha158 set spans (i) basic price–volume indicators (OHLC, volume, VWAP); (ii) trend indicators (moving averages over 5–60 days and K-line patterns); (iii) momentum indicators (Rate of Change, Raw Stochastic Value); and (iv) volatility and correlation measures (standard deviation, range extremes, Beta). Minor missing values from trading halts are linearly interpolated, and a sliding window segments the data into sequential inputs for the BIGRU.

4.2 Model Configuration and Hyperparameters

Table 1 reports the HFSLS/BIGRU configuration together with the PSO search space and swarm parameters.

Table 1: Model configuration: HFSLs/BIGRU hyperparameters (left) and the PSO search space and swarm parameters (right).

Category	Parameter	Value	Type	Parameter	Value
HFSLs	Sub-models (K)	3	Search Sp.	Hidden Units	[32, 96]
	Selection Ratios	[1.0, 0.9, 0.8]		Dropout Rate	[0.1, 0.4]
	Sampling Bins	3		Learning Rate	[0.0005, 0.005]
	Window Size	30			
BIGRU	Optimizer	Adam	Swarm Cfg.	No. of Particles	10
	Loss Function	MSE		Max Iterations	25
	Hidden Layers	Bidir.		Inertia Wt. (ω)	0.7
	Activation	ReLU		Cognitive (C_1)	1.5
	Training Epochs	30		Social (C_2)	1.5
	Batch Size	32			

4.3 Evaluation Metrics

Three regression metrics evaluate predictive performance: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2),

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (6)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (7)$$

where y_i is the actual value, $\hat{y}_i$ the predicted value, and $\bar{y}$ the mean. RMSE penalizes large deviations more heavily, MAE is robust to outliers, and R^2 measures goodness of fit.

5 Prediction Results

5.1 TAIEX and Ablation Study

On the TAIEX, the model’s predictions align closely with actual closing prices, particularly during the high-volatility periods of 2024–2025; only a slight lag appears during the sharp April 2025 correction, after which the trajectory remains stable. Table 2 reports the ablation results for the TAIEX. HFSLs-BIGRU significantly outperforms the baseline BIGRU, improving R^2 by 2.392 and reducing MSE by 97.1%, confirming that the HFSLs module plays the critical role. In contrast, PSO-BIGRU performs worse than the baseline (R^2 decreases by 1.661 and MSE increases by 69.6%), suggesting that applying PSO alone in a high-dimensional feature space may destabilize learning. Adding PSO on top of HFSLs yields only a marginal improvement (0.002 in R^2), so the full model achieves a total improvement of 2.394 in R^2 and a 97.1% reduction in MSE relative to the baseline.

Table 2: Ablation study results for TAIEX prediction.

Model	R^2	MSE	RMSE	MAE	MAPE	MAE-Orig.
BIGRU	−1.457	0.069	0.264	0.252	7.993	1,784.400
PSO-BIGRU	−3.118	0.117	0.341	0.329	10.451	2,331.433
HFSLs-BIGRU	0.935	0.002	0.043	0.030	0.985	215.344
HFSLs-PSO-BIGRU	0.937	0.002	0.042	0.032	1.040	226.928

Note: MAE-Original is the error rescaled back to the original units to provide practical interpretability.

5.2 Cross-Sector Generalization

Table 3 reports performance across the four stocks. The baseline BIGRU and PSO-BIGRU attain unstable average R^2 values of 0.088 ($\sigma = 0.656$) and -0.768 ($\sigma = 2.045$), whereas HFSLs-BIGRU reaches 0.926 ($\sigma = 0.029$) and the full model 0.902 ($\sigma = 0.096$), the latter winning four of five targets—evidence of robust generalization.

A visual comparison of predicted versus actual closing prices reinforces these statistics: the baseline drifts and fails to capture the 2024 TSMC uptrend and responds unstably to abrupt moves in the high-volatility Evergreen and MediaTek, whereas the proposed model tracks all four price trajectories closely and smoothly across different market regimes.

Table 3: Performance comparison of model generalization across stocks. Models: B = BIGRU, P = PSO-BIGRU, H = HFSLs-BIGRU, F = HFSLs-PSO-BIGRU (full). Stocks: TSMC, EVG = Evergreen, MTK = MediaTek, UPE = Uni-President.

Metric	Stk	B	P	H	F
R^2	TSMC	-0.703	-3.893	0.952	0.959
	EVG	0.266	0.115	0.894	0.731
	MTK	0.544	-3.101	0.905	0.928
	UPE	0.837	0.608	0.945	0.953
MSE	TSMC	0.084	0.243	0.002	0.002
	EVG	0.033	0.008	0.005	0.012
	MTK	0.020	0.174	0.004	0.003
	UPE	0.005	0.013	0.001	0.001
RMSE	TSMC	0.290	0.493	0.049	0.045
	EVG	0.183	0.090	0.068	0.109
	MTK	0.142	0.417	0.063	0.055
	UPE	0.072	0.114	0.034	0.039

Metric	Stk	B	P	H	F
MAE	TSMC	0.273	0.476	0.039	0.035
	EVG	0.150	0.070	0.049	0.088
	MTK	0.107	0.400	0.048	0.043
	UPE	0.055	0.096	0.025	0.028
MAPE	TSMC	12.354	21.543	1.838	1.640
	EVG	7.071	3.486	2.385	4.256
	MTK	4.918	17.849	2.179	1.953
	UPE	1.358	2.355	0.620	0.703
MAE-Orig.	TSMC	124.873	218.034	18.036	16.170
	EVG	15.012	7.031	4.884	8.774
	MTK	63.653	237.734	28.789	25.803
	UPE	3.715	6.420	1.670	1.902

6 Trading Strategy and Backtesting Results

6.1 Overlapping Trade Problem and Capital Allocation

The model predicts the price at $T + 11$. Generating a signal every day creates overlapping positions (one opened at t is still held when $t + 1$ triggers another), while a non-overlapping “open–hold–close” scheme executes only about 286/11 trades, wasting signals and capital. We therefore adopt a *capital-allocation* method: the initial capital (\$1,000,000) is split into 11 equal sub-portfolios, and each day one sub-portfolio opens a position held for 11 days, so up to 11 time-diversified positions coexist. The portfolio return averages the individual trades,

$$R_{\text{portfolio}} = \frac{1}{N} \sum_{i=1}^N R_i, \quad (8)$$

where R_i is the return of the i -th trade and N the total number of trades.

6.2 Backtesting Methodology

The backtesting period spans April 1, 2024 to August 18, 2025 (383 valid observations per stock). A symmetric long–short strategy is adopted: the model predicts the closing price at $T + 11$, opens a long position if it exceeds the current price and a short position otherwise, and closes after 11 trading days—matching the prediction horizon.

6.3 Backtesting Results: BIGRU versus HFSLs-PSO-BIGRU

Table 4 compares the baseline BIGRU and the full model. The baseline is weak and unstable—TSMC even yields a -23.57% strategy return, below its buy-and-hold, with several profit factors near one. The full model instead exceeds an 86% total win rate on every stock and delivers strategy returns far above buy-and-hold—598.81% (Evergreen), 563.71% (MediaTek), and 356.86% (TSMC)—with the Evergreen profit

factor reaching 29.53. The top row of Fig. 2 confirms the proposed model consistently produces the highest cumulative returns, staying on an upward trajectory even when passive holding stagnates.

Table 4: Backtesting performance: baseline BIGRU (B) versus the full HFSLs-PSO-BIGRU model (F), April 2024–August 2025.

Metric	TSMC		Evergreen		MediaTek		Uni-President	
	B	F	B	F	B	F	B	F
Total Trades	383	383	383	383	383	383	383	383
Long Trades	42	247	95	182	174	195	154	174
Short Trades	341	146	288	211	209	198	229	219
Long Win (%)	100	93.93	77.90	95.60	81.61	96.92	80.52	87.93
Short Win (%)	35.19	78.77	54.86	84.83	61.72	78.28	71.62	84.48
Total Win (%)	42.30	88.30	60.57	89.82	70.76	87.53	75.20	86.01
Strategy Ret. (%)	−23.57	356.86	107.93	598.81	217.96	563.71	27.07	42.50
Buy-Hold (%)	101.20	99.25	26.77	21.74	46.91	48.39	−4.00	−5.33
Excess Ret. (%)	−124.77	257.61	81.15	577.08	171.05	515.32	31.08	47.83
Avg. Ret./Trade (%)	−0.77	4.34	2.13	5.59	3.38	5.44	0.69	1.00
Profit Factor	0.75	20.45	2.03	29.53	3.45	25.21	2.77	6.60

Note: Excess Return is Strategy Return minus Buy-and-Hold Return. Profit Factor is the ratio of total profits to total losses.

6.4 Comparative Analysis against the TAIEX

The strategy is also benchmarked against the TAIEX (Table 5), using the 286 observations with both stock and index data for a consistent time frame. Strategy returns significantly outperform the TAIEX for TSMC, Evergreen Marine, and MediaTek, and exceed it by a smaller margin for Uni-President. As shown in the bottom row of Fig. 2, the framework keeps a steeper cumulative-return trajectory across momentum-driven stocks and, for Uni-President, a more stable path during the early-2025 downturn, indicating downside resilience.

Table 5: Performance comparison of the HFSLs-PSO-BIGRU model against the TAIEX.

Performance Metric	TSMC	Evergreen	MediaTek	Uni-President
Total Trades	286	286	286	286
Long Trades	180	133	140	127
Short Trades	106	153	146	159
Long Win Rate (%)	76.667	83.459	80.000	30.709
Short Win Rate (%)	76.415	64.052	67.123	54.088
Total Win Rate (%)	76.573	73.077	73.427	43.706
Strategy Return (%)	203.137	304.597	289.862	30.829
TAIEX Return (%)	28.286	28.286	28.286	28.286
Excess Return (%)	174.851	276.311	261.576	2.543
Profit Factor	9.885	6.526	6.084	1.046

Note: Excess Return is Strategy Return minus TAIEX Return. Profit Factor is the ratio of total profits to total losses.

7 Conclusion

This study proposes the HFSLs-PSO-BIGRU hybrid forecasting model, integrating hierarchical feature selection, particle swarm optimization, and bidirectional gated recurrent units to address high dimensionality, nonlinearity, and long-term dependencies in financial time series. Ablation experiments identify HFSLs as the dominant driver of performance, while PSO is beneficial only in combination with feature selection. Translated into a long–short strategy, the model consistently outperforms the market: for TSMC, Evergreen Marine, and MediaTek the strategy achieves returns substantially higher than the TAIEX and remains stable during volatility. Validated across technology, shipping, and consumer-goods sectors, the framework demonstrates robust cross-market and cross-sector generalization, offering a methodological foundation for

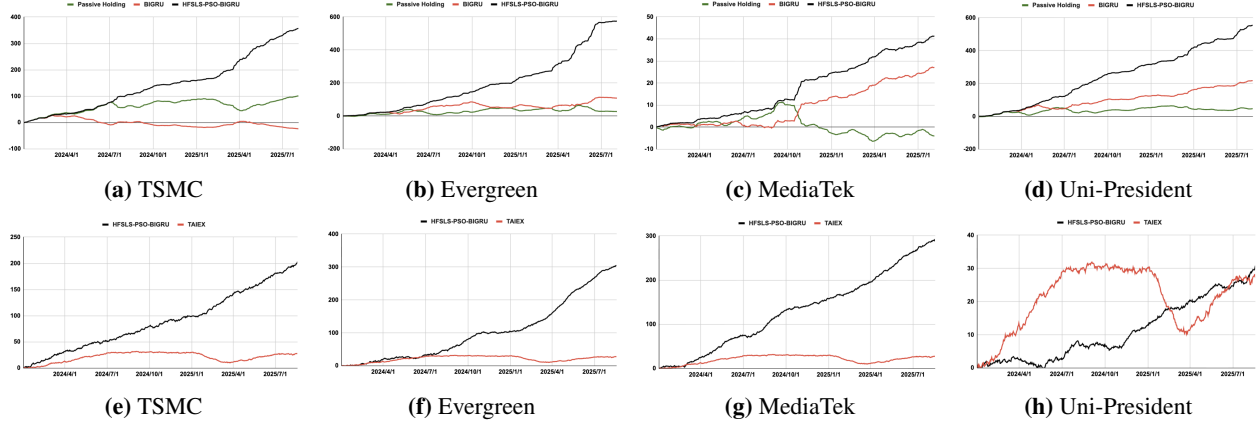


Figure 2: Cumulative return curves of the HFSLs-PSO-BIGRU strategy. Top row: versus the BIGRU baseline and passive holding. Bottom row: versus the TAIEX benchmark.

quantitative trading. Future work will incorporate transaction-cost modeling, attention-based architectures, and live-trading validation.

References

- Bailey, D. H., Borwein, J. M., López de Prado, M., and Zhu, Q. J. (2014). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39–69.
- Bengio, Y., Simard, P., and Frasconi, P. (1994). Learning long-term dependencies with gradient descent is difficult. *IEEE Transactions on Neural Networks*, 5(2):157–166.
- Bergstra, J. and Bengio, Y. (2012). Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13:281–305.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.
- Box, G. E. P. and Jenkins, G. M. (1970). *Time Series Analysis: Forecasting and Control*. Holden-Day, San Francisco, CA, USA.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.
- Chen, S. and Tang, G. (2025). A layered stock prediction model based on novel feature selection and model parameter optimization. *The Journal of Supercomputing*, 81(13):1256.
- Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., and Bengio, Y. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078*.
- Engle, R. F. (1982). Autoregressive conditional heteroskedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007.
- Fama, E. F. and French, K. R. (1992). The cross-section of expected stock returns. *The Journal of Finance*, 47(2):427–465.
- Fischer, T. and Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669.
- Goodfellow, I., Bengio, Y., and Courville, A. (2016). *Deep Learning*. MIT Press, Cambridge, MA, USA.
- Greff, K., Srivastava, R. K., Koutník, J., Steunebrink, B. R., and Schmidhuber, J. (2017). LSTM: A search space odyssey. *IEEE Transactions on Neural Networks and Learning Systems*, 28(10):2222–2232.
- Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273.
- Guyon, I. and Elisseeff, A. (2003). An introduction to variable and feature selection. *Journal of Machine Learning Research*, 3:1157–1182.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ... and the cross-section of expected returns. *The Review of Financial Studies*, 29(1):5–68.
- Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning*. Springer, New York, NY, USA, 2nd edition.
- Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8):1735–1780.
- Kelly, B. T., Pruitt, S., and Su, Y. (2019). Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501–524.
- Kennedy, J. and Eberhart, R. (1995). Particle swarm optimization. In *Proceedings of the IEEE International Conference on Neural Networks*, volume 4, pages 1942–1948, Perth, Australia.
- Krauss, C., Do, X. A., and Huck, N. (2017). Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500. *European Journal of Operational Research*, 259(2):689–702.
- LeCun, Y., Bengio, Y., and Hinton, G. (2015). Deep learning. *Nature*, 521(7553):436–444.
- Li, J., Cheng, K., Wang, S., Morstatter, F., Trevino, R. P., Tang, J., and Liu, H. (2017). Feature selection: A data perspective. *ACM Computing Surveys*, 50(6):1–45.
- Lim, B., Arık, S. Ö., Loeff, N., and Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley, Hoboken, NJ, USA.
- Malkiel, B. G. (2003). The efficient market hypothesis and its critics. *Journal of Economic Perspectives*, 17(1):59–82.
- Schuster, M. and Paliwal, K. K. (1997). Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing*, 45(11):2673–2681.
- Vapnik, V. N. (1995). *The Nature of Statistical Learning Theory*. Springer, New York, NY, USA.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. (2017). Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS)*, pages 5998–6008, Long Beach, CA, USA.